Automated Network Request Management in ServiceNow

FINAL PROJECT REPORT

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Platform	ServiceNow
Domain	IT Service Management (ITSM) / Network Operations
Project Type	Academic / Industry Simulation Project
Tools Used	ServiceNow Flow Designer, Service Catalog, Workflow Editor, Form Designer, Notification Engine, SLA Management

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1. Introduction

1.1 Project Overview

The Automated Network Request Management System is a comprehensive, enterprise-grade solution developed on the ServiceNow platform. It digitizes and automates the full lifecycle of network service requests — from submission and categorization through multi-stage approval, fulfillment, and closure — within a centralized, auditable, ITSM-compliant framework. The system leverages ServiceNow's Service Catalog for structured intake, Flow Designer for workflow automation, and its Notification Engine for real-time stakeholder communication.

1.2 Problem Statement

Organizations operating complex network infrastructures face recurring challenges when relying on manual, unstructured request channels. Key pain points include:

- Email and phone-based request intake with no traceability, causing lost or overlooked requests.
- No standardized intake process, leading to incomplete data and repeated clarification cycles.
- Informal approval chains creating compliance risks and audit failures.
- No unified engineer task queue, causing confusion over task ownership and priority.
- Zero real-time visibility for requesters into their submission status.
- Manual SLA tracking resulting in frequent breaches and frustrated end users.
- Inability to generate meaningful reporting or trend analysis from scattered request data.

1.3 Objectives

- Deploy a structured Service Catalog with 8 categorized network request types.
- Build dynamic, validated forms capturing all required data for each request type.
- Implement role-based access control ensuring appropriate visibility and actions for all user groups.
- Configure multi-stage approval workflows with conditional risk-based routing.
- Integrate automated email notifications at every lifecycle stage.
- Establish automated task assignment and SLA tracking with escalation mechanisms.
- Validate end-to-end through comprehensive testing and UAT.
- Deploy to production with full training and post-deployment support.

2. Literature Review

2.1 IT Service Management and ITSM Frameworks

IT Service Management (ITSM) encompasses the processes, policies, and procedures organizations use to design, deliver, manage, and improve IT services. The ITIL 4 framework — the most widely adopted global ITSM standard — provides guidance across service strategy, design, transition, operation, and continual improvement. Request Management is a key ITIL process that handles standard, pre-approved service requests separately from incidents or changes.

2.2 ServiceNow as an Enterprise ITSM Platform

ServiceNow is an industry-leading cloud-based ITSM platform offering unified modules for Incident, Problem, Change, Service Catalog, and Request Management. Its Service Catalog enables consumer-style service browsing and structured form-based request submission. Flow Designer provides no-code and low-code workflow automation, while native role-based access control ensures secure, appropriate data visibility across all user groups.

2.3 Network Request Management — Research Context

Enterprise IT operations research consistently identifies network request management as one of the most process-intensive and error-prone service delivery areas. Organizations without automation experience significantly higher MTTR, lower customer satisfaction, and elevated non-compliance risk. Automation of network request workflows has been demonstrated to reduce MTTR by up to 60%, eliminate manual data entry errors, and substantially improve SLA compliance.

3. Ideation Phase — Empathy & Problem Discovery

3.1 User Personas Identified

Persona	Primary Need
IT Staff / Requester	Needs to submit network requests quickly and track status without chasing IT teams.
Line Manager / Approver	Needs structured, contextual approval requests with deadline visibility and delegation support.
Network Engineer	Needs clearly scoped, fully detailed fulfillment tasks routed directly to their queue.
IT Administrator	Needs centralized visibility, audit trails, SLA dashboards, and compliance reporting.

3.2 Empathy Map Key Insights

- All user types consistently expressed the need for a centralized, structured portal as the single intake point.
- The dominant emotional state across personas is uncertainty — with no visibility into request status or ownership.
- Automated routing, notifications, and SLA escalation are critical to user satisfaction and compliance.
- Dynamic, validated intake forms are essential to eliminate incomplete submissions and clarification cycles.

3.3 Problem Statement Summary (All Personas)

PS ID	Persona	Core Problem	Emotional Impact
PS-1	IT Requester	No centralized portal; manual email submission; no status visibility.	Frustrated, anxious, uncertain.
PS-2	Line Manager	No structured approval requests; no delegation support for absences.	Disappointed, overwhelmed.
PS-3	Network Engineer	Tasks arrive informally with missing details; no structured queue.	Overwhelmed, SLA-concerned.
PS-4	IT Administrator	No audit trail; scattered data; manual reporting impossible.	Compliance-concerned.

4. Requirement Analysis

4.1 Customer Journey Summary

The end-to-end user journey comprises four stages: Discovery & Intake (finding the catalog item), Submission & Validation (completing the dynamic form), Approval & Governance (multi-stage workflow routing), and Fulfillment & Resolution (task execution and closure notification). Each stage was mapped for user actions, touchpoints, emotions, pain points, and improvement opportunities.

4.2 Key Functional Requirements

- FR1: User Authentication & Role Management — sys_user authentication with explicit role assignment (requester, approver, engineer).
- FR2: Service Catalog & Intake — 8 distinct catalog items with dynamic, regex-validated forms.
- FR3: Multi-Stage Approval Engine — conditional routing through Line Manager and Security Team with mandatory rejection comments.
- FR4: Queue Management & Task Assignment — automated sc_task generation and Assignment Rule routing.
- FR5: Workflow & Notification Automation — Flow Designer-driven lifecycle with email notifications at every stage.
- FR6: SLA Tracking & Escalation — automated SLA clocks with escalation at 50%, 75%, and 100% breach thresholds.

4.3 Key Non-Functional Requirements

- Usability: Consumer-grade Service Portal with tooltips for non-technical users.
- Security: ACL-enforced role-based data access with field-level restrictions.
- Reliability: Immutable, date-stamped audit entries for all state changes.
- Performance: Sub-2-second portal form load times; minimal backend processing latency.
- Availability: 99.9% uptime matching the ServiceNow cloud instance SLA.
- Scalability: Schema supports growing request volumes and catalog expansion without redesign.

5. Project Design Phase

5.1 Problem–Solution Fit

The solution directly addresses all four persona problem statements: the structured Service Catalog eliminates ad-hoc intake; Flow Designer removes manual routing; automated notifications close the visibility gap; ACLs enforce compliance; and the SLA engine prevents breach-related frustration.

5.2 Solution Architecture — Four Processing Layers

Presentation: ServiceNow Service Portal (requesters) + Native ITSM Workspace (engineers/approvers).

Application & Security: UI Policies + Client Scripts for form validation; ACLs for role-based data access.

Automation Engine: Flow Designer for approval routing, task generation, SLA management, and notifications.

Storage: sc_req_item, sysapproval_approver, sc_task, and custom u_network_database tables.

6. Project Planning Phase — Agile Sprint Plan

Sprint	Epic	Story No.	Story Points	Timeline
Sprint-1	Project Setup & Portal Intake	USN1	8	22–23 June 2026
Sprint-2	Form Setup & Validation Logic	USN2	9	24–25 June 2026
Sprint-3	Core Workflows & SLA Management	USN3	8	26–27 June 2026
Sprint-4	End-to-End Testing & Deployment	USN4	5	28 June 2026

Total: 30 story points delivered across 7 days. Average velocity: 4.29 story points per day. All 4 sprints completed on schedule with UAT sign-off achieved on Day 7.

7. Project Development Phase — Implementation Milestones

Milestone 1: Application Setup & Portal Initialization

Established Service Catalog taxonomy with 8 network request items under the Network Requests category. Configured Service Portal accessibility for all user roles. Initialized the development environment on a ServiceNow personal developer instance.

Milestone 2: Database & Schema Configuration

Created the custom u_network_database table with columns: u_request_number, u_requested_for, u_device_type, u_work_status, and u_updates. Validated table dictionary structure and field indexing for operational query performance.

Milestone 3: Role-Based Access Control & Security Matrix

Provisioned test user records and assigned them to sn_network_requester, sn_network_approver, and sn_network_engineer role groups. Configured ACL rules governing create, read, write, and delete access on all operational tables. Validated all security constraints.

Milestone 4: User Interface & Dynamic Form Engineering

Engineered dynamic catalog forms for all 8 request types. Applied regex Client Scripts for IP/VLAN/port validation. Configured UI Policies for conditional field visibility. All forms enforce mandatory field rules blocking incomplete submissions.

Milestone 5: Workflow Orchestration & Notification Engine

Built Flow Designer workflows for all 8 catalog items with risk-based conditional routing. Configured SLA management with automated escalation. Deployed email notification templates for all lifecycle transitions. Validated end-to-end workflow execution through full scenario testing.

8. Testing & Quality Assurance

8.1 Testing Strategy

A comprehensive four-phase testing strategy was employed: unit testing (individual forms and validations), integration testing (workflow-to-notification linkages), end-to-end testing (full lifecycle scenarios), and User Acceptance Testing (UAT) with multi-stakeholder participation. Negative testing validated error handling, security enforcement, and SLA escalation logic.

8.2 Key Test Scenarios

- Successful catalog item submission with valid data — confirmed workflow trigger and submission notification.
- Submission with missing mandatory fields — confirmed validation blocks form submission.
- Multi-stage approval routing — confirmed correct approver sequencing at each stage.
- Request rejection at each approval stage — confirmed notifications and rejection audit records.
- Fulfillment task creation and assignment — confirmed sc_task generation and queue routing.
- SLA breach simulation — confirmed escalation notifications at 50%, 75%, and 100% thresholds.
- Unauthorized access attempt — confirmed ACL-based role restrictions enforced correctly.
- End-to-end closure — confirmed request and task closure sequence and completion notification.

8.3 UAT Outcomes

UAT was conducted with representatives from IT operations, network engineering, line management, and end-user communities. All critical and high-severity defects were resolved prior to production deployment. UAT sign-off obtained from all stakeholder groups.

9. Results & Outcomes

9.1 Key Achievements

- Fully operational Service Catalog with 8 categorized network request types accessible via the Service Portal.
- Dynamic, validated forms for all catalog items with conditional logic and mandatory field enforcement.
- Automated multi-stage approval workflows with conditional routing and email notifications at every stage.
- SLA monitoring with automated escalation at 50%, 75%, and 100% breach thresholds.
- Automated fulfillment task creation and routing, eliminating manual task assignment.
- Complete, immutable audit trail for all requests, approvals, and fulfillment activities.
- Successful production deployment with full UAT sign-off.
- Training delivered to all user groups with comprehensive documentation.

9.2 Benefits Realized

- Eliminated email and phone-based request intake, replaced with structured, trackable catalog process.
- Reduced incomplete submissions through mandatory field enforcement and dynamic form guidance.
- Improved real-time visibility for all stakeholders through status tracking and automated notifications.
- Enhanced compliance through enforced approval chains and immutable audit logging.
- Reduced administrative burden through workflow automation and automated task creation.
- Improved SLA adherence through automated monitoring and escalation.
- Standardized request data enabling meaningful reporting and trend analysis.

10. Conclusion

10.1 Summary

The Automated Network Request Management System in ServiceNow represents a meaningful and measurable advancement in enterprise network service delivery. By replacing informal, fragmented request channels with a structured, automated, and ITSM-aligned system, the project delivers improvements in efficiency, compliance, visibility, and user experience across all stakeholder groups. The five-milestone development structure provided a logical, validated pathway from initial catalog design through workflow automation and successful production deployment.

10.2 Future Enhancements

- Integration with Cisco DNA Center and SolarWinds for automated network configuration execution upon task approval.
- Self-service knowledge base within the Service Portal to guide users in selecting the correct catalog item.
- Real-time analytics dashboard providing visibility into request volumes, SLA performance, and approval cycle times.
- Extended catalog covering cloud resource provisioning and server infrastructure requests.
- ServiceNow Virtual Agent chatbot for conversational request submission and real-time status inquiry.

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